

M NITHIN

GenAI / ML Engineer | RAG Systems | Python Backend Development

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SUMMARY

GenAI/ML Engineer with 2+ years of production experience building **RAG-based LLM applications**, scalable **Python/FastAPI** backend systems, and multi-agent orchestration pipelines for healthcare AI. Delivered **sub-1.5s query latency**, **92% retrieval accuracy**, and a **60-65% improvement** in retrieval/ingestion efficiency on production clinical systems. Experienced with vector databases (ChromaDB, FAISS), LangChain, LlamaIndex, on-premise LLM inference (Ollama), and containerized deployment (Docker).

SKILLS

Programming Languages : Python

Frameworks & Libraries : FastAPI, Flask, PyTorch, scikit-learn, Transformers, React (Basic)

GenAI / LLM : RAG, LangChain, LlamaIndex, Agentic AI, Multi-Agent Orchestration, Prompt Engineering, Embeddings, Semantic Search, Hugging Face

Backend Development : REST APIs, Model Deployment, Async Processing

Databases : MySQL, Vector Databases (ChromaDB, FAISS)

Data Visualization : Matplotlib, Seaborn, Plotly

Tools & Platforms : Git, Bitbucket, JIRA, Postman, Docker

Cloud & DevOps : AWS / GCP (Basic), Docker

WORK EXPERIENCE

DALVKOT UTILITY ENTERPRISES LIMITED

October 2024 - Present

Software Developer

Bengaluru

Project: Medical GenAI – RAG-Based Clinical Question Answering System

Tech Stack: Python, FastAPI, LangChain, ChromaDB, Hugging Face Transformers, Ollama, React

- Designed and implemented a RAG-based clinical question answering system using ChromaDB and Hugging Face embeddings, with PyPDF/OCR-based ingestion to structure unstructured medical documents.
- Implemented semantic search using vector similarity (MiniLM embeddings) with relevance filtering and fallback mechanisms to reduce hallucinations and ensure reliable, clinically safe responses.
- Deployed Ollama for on-premise LLM inference to meet clinical data privacy and compliance requirements.
- Built a FastAPI-based backend integrated with a clinician-facing dashboard, enabling patient record queries with <1.5s latency.
- Improved clinical information retrieval efficiency by 60%, reduced irrelevant responses by 40%, achieving 92% retrieval accuracy.

Project: AI-Powered Dynamic Discharge Summary Generation System

Tech Stack: Python, FastAPI, LangChain, MySQL, Ollama / OpenAI API, React

- Developed a multi-agent agentic AI system in Python for clinical document generation, ingesting encounter data via SQL stored procedures and normalizing patient records.
- Engineered specialized agent pipelines for vital trend analysis, clinical entity extraction, and diagnosis/procedure summarization — improving completeness and structure of discharge summaries.
- Built a main summary orchestration layer with optional Ollama-based rewriting, producing standardized JSON outputs for full and dynamic header-based summaries.
- Implemented a frontend-integrated workflow with a dynamic header agent generating validated, section-wise summaries, plus an OpenAI/Ollama model switcher for flexible cloud/on-premise deployment.

EDUCATION

SREE RAMA ENGINEERING COLLEGE, TIRUPATI

2019-2022

B.Tech, Electronics and Communication Engineering

CGPA : 6.67